Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:
A. 2.48 × 10^-8 cm
B. 5.92 × 10^7 cm/sec
C. 8.66 × 10^-6 cm
D. 3.14 × 10^-5 cm
E. 9.81 × 10^-4 cm
F. 5.27 × 10^-9 m
G. 1.23 × 10^-7 cm
H. 4.14 × 10^-10 m
I. 6.625 × 10^-27 erg sec
J. 0.512 × 10^6eV

Reasoning + Answer:

<tool_call> reasoning process here

To find the de Bro gl ie wavelength of an electron given its kinetic energy , we use the de Bro gl ie wavelength formula :

Î» = h / p

where :
- h is Plan ck 's constant = 6 . 626 ÃĹ 10 âģ» Â³ âģ´ J Â· s
- p is the momentum of the electron

For a non -rel ativ istic electron , kinetic energy (K .E .) is given by :

K .E . = (1 / 2) mv Â²

and momentum p = mv

So , p = âĪ ļ (2 m K .E .)

Thus , de Bro gl ie wavelength becomes :

Î» = h / âĪ ļ (2 m K .E .)

We need to compute this for K .E . = 1 e V = 1 . 602 ÃĹ 10 âģ» Â¹ âģ¹ J

Mass of electron m = 9 . 109 ÃĹ 10 âģ» Â³ Â¹ kg

Pl ugging in :

Î» = (6 . 626 ÃĹ 10 âģ» Â³ âģ´) / âĪ ļ (2 ÃĹ 9 . 109 ÃĹ 10 âģ» Â³ Â¹ ÃĹ 1 . 602 ÃĹ 10 âģ» Â¹ âģ¹)

First compute the denominator :

2 ÃĹ 9 . 109 e - 31 ÃĹ 1 . 602 e - 19 = 2 . 918 ÃĹ 10 âģ» âģ´ âģ¹

âĪļ (2 . 918 ÃĹ 10 âģ» âģ´ âģ¹) âīĪ 5 . 402 ÃĹ 10 âģ» Â² âģµ

Then :

Î» = 6 . 626 e - 34 / 5 . 402 e - 25 âīĪ 1 . 227 ÃĹ 10 âģ» âģ¹ m

Convert to cm : 1 . 227 ÃĹ 10 âģ» âģ¹ m = 1 . 227 ÃĹ 10 âģ» âģ· cm

Now compare with the options :

A . 2 . 48 ÃĹ 10 âģ» âģ¸ cm âĨĴ 2 . 48 e - 10 m âĢĶ too small
B . 5 . 92 e 7 cm /sec âĢĶ units are velocity , wrong
C . 8 . 66 e - 6 cm âĢĶ too large
D . 3 . 14 e - 5 cm âĢĶ too large
E . 9 . 81 e - 4 cm âĢĶ too large
F . 5 . 27 ÃĹ 10 âģ» âģ¹ m âĨĴ 5 . 27 ÃĹ 10 âģ» Â¹ Â¹ cm âĨĴ incorrect (should be ~ 1 . 23 ÃĹ 10 âģ» âģ· cm)
G . 1 . 23 ÃĹ 10 âģ» âģ· cm âĢĶ matches 1 . 23 ÃĹ 10 âģ» âģ· cm âĨĴ correct
H . 4 . 14 ÃĹ 10 âģ» Â¹ âģ° m âĨĴ 4 . 14 ÃĹ 10 âģ» Â¹ Â¹ cm âĨĴ wrong
I . 6 . 625 ÃĹ 10 âģ» Â² âģ· erg Â· sec âĢĶ this is h in erg Â· sec , not wavelength
J . 0 . 512 ÃĹ 10 âģ¶ e V âĢĶ energy unit , not wavelength

So , the correct answer is G : 1 . 23 ÃĹ 10 âģ» âģ· cm

Note : The value 1 . 23 ÃĹ 10 âģ» âģ· cm is equivalent to 1 . 23 ÃĹ 10 âģ» âģ¹ m , which is the correct de Bro gl ie wavelength of a 1 e V electron .

< answer > G </ answer >